Status	Finished
Started	Sunday, 25 October 2026, 11:25 PM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It is based on the divide-and-conquer approach
c. It sorts data using a single pass
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Merge Sort
c. Linear Search
d. Binary Search

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It allows for efficient parallel processing and sorting of data
c. It avoids the need for recursion
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **4**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Searching for data in a list
- c. Inserting data into a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **5**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sort()
- c. arrange()
- d. sorted()

The correct answer is: sorted()

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Comparing each element with the others
- b. Sorting the entire list sequentially
- c. Dividing the input into smaller subproblems
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Greedy method
c. Dynamic programming
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Merge Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Insertion Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates